Classifying Rational and Irrational Numbers
Answer Key
Grade 8 / Pre-Algebra

Quick Answers

1. $\frac{7}{20}$
2. 11
3. (a) lower whole number = 7, (a) upper whole number = 8, (b) nearest hundredth = 7.35, —
4. the fraction in lowest terms = $\frac{2}{3}$, —
5. (a) the value = 13, —
6. (a) the comparison = <, —
7. (a) lowest terms = $\frac{3}{4}$, (b) decimal form = 0.75
8. $\frac{3}{2}$
9. $\sqrt{7}$, $\sqrt{9}$, π, $\frac{10}{3}$
10. (a) the fraction = $1 \cdot \frac{1}{6}$, —

What is verified

5 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 3, 4, 5, 6, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra
8.NS.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 18 tagged checks.

1. Write 0.35 as a fraction of two integers in lowest terms.

Solution: The decimal terminates in the hundredths place, so it is $\dfrac{35}{100}$. Dividing numerator and denominator by their common factor five gives lowest terms, which proves the number is rational. $\boxed{\frac{7}{20}}$

2. Evaluate $\sqrt{121}$, and write the value as a fraction of two integers.

Solution: The radicand is a perfect square, since $11 \times 11 = 121$, so the root is a whole number. Every whole number is itself over one, so the value is rational. $\boxed{11}$

3. $\sqrt{54}$ cannot be written as a fraction, so locate it instead. (a) Between which whole numbers? (b) To the nearest hundredth? (c) Mark it.

Solution (a): Trap the radicand between neighbouring perfect squares: $7^2 = 49$ and $8^2 = 64$, and $49 < 54 < 64$. Taking roots keeps the order, so the value lies between $\boxed{7}$ and $\boxed{8}$.

Solution (b): Squeeze with hundredths: $7.34^2 = 53.8756$ and $7.35^2 = 54.0225$, so the root is between them and rounds up. $\boxed{7.35}$

Solution (c) — instructor-judged. The dot belongs between the whole number ticks found in part (a), noticeably past the middle of that interval (the value is about a third of the way past the lower tick), and labelled with the root symbol or the rounded decimal. A dot left of the midpoint, or sitting on a tick, is not correct.

4. Show that the repeating decimal $0.\overline{6}$ is rational, using $x = 0.\overline{6}$ and the subtract-the-copy method.

Solution:

$$\begin{aligned} x &= 0.666\ldots \\ 10x &= 6.666\ldots \\ 10x - x &= 6.666\ldots - 0.666\ldots \\ 9x &= 6 \end{aligned}$$

The repeating tails cancel exactly, leaving a whole number on the right. Dividing by nine and reducing gives a ratio of two integers, so the number is rational. $\boxed{x = \frac{2}{3}}$

5. Consider $\sqrt{169}$. (a) Evaluate it. (b) Explain how the value shows the number is rational.

Solution (a): $13 \times 13 = 169$, so the radicand is a perfect square. $\boxed{13}$

Solution (b) — instructor-judged. A correct sentence links the computed whole number to the definition: the root came out a whole number, and any whole number can be written over one, so the value is a ratio of two integers and therefore rational. Answers that only restate the value, or that claim the radical sign forces irrationality, do not earn credit.

6. One of π and $\dfrac{22}{7}$ is irrational and one is rational. (a) Write $<$, $>$, or $=$ between them. (b) Mark both on the number line.

Solution (a): Write out enough digits to separate them: $\pi = 3.14159\ldots$ while $\dfrac{22}{7} = 3.142857\ldots$, and the two decimals first differ in the thousandths place, where one is smaller. $\boxed{<}$

Solution (b) — instructor-judged. Two labelled dots, both between the ticks at three and one tenth and at three and two tenths, with the fraction's dot to the *right* of the one for the circle constant. Credit depends on the left-to-right order being right, not on perfect placement.

7. Consider $\dfrac{9}{12}$. (a) Write it in lowest terms. (b) Write it as a decimal.

Solution (a): Both parts share a factor of three, so dividing top and bottom by three reduces the fraction. $\boxed{\frac{3}{4}}$

Solution (b): Dividing the reduced numerator by the reduced denominator terminates after two places. $\boxed{0.75}$

8. Evaluate $\sqrt{\dfrac{9}{4}}$ and write the value as a fraction of two integers in lowest terms.

Solution: The root of a fraction is the root of the top over the root of the bottom, and both parts are perfect squares. The result is already in lowest terms, so this root is rational even though it is not a whole number. $\boxed{\frac{3}{2}}$

9. Write $\sqrt{7}$, $\sqrt{9}$, π, $\dfrac{10}{3}$ in order from least to greatest.

Solution: Convert every entry to a decimal before comparing: $\sqrt{7} = 2.6457\ldots$, $\sqrt{9} = 3$, $\pi = 3.14159\ldots$, and $\dfrac{10}{3} = 3.333\ldots$. Two of these are irrational and two are rational, but ordering only needs the decimals.

$$\boxed{\sqrt{7} \; < \; \sqrt{9} \; < \; \pi \; < \; \tfrac{10}{3}}$$

10. Challenge. Tomas says the digits of $0.1\overline{6}$ never stop, so it must be irrational. (a) Write it as a fraction. (b) Explain what is wrong with his reason.

Solution (a): Only the last digit repeats, so shift twice and subtract.

$$\begin{aligned} x &= 0.1666\ldots \\ 10x &= 1.666\ldots \\ 100x &= 16.666\ldots \\ 100x - 10x &= 16.666\ldots - 1.666\ldots \\ 90x &= 15 \end{aligned}$$

Dividing by ninety and reducing gives a ratio of two integers. $\boxed{x = \frac{1}{6}}$

Solution (b) — instructor-judged. Tomas has confused "never ends" with "never repeats". A decimal whose digits eventually repeat a fixed block forever is rational — part (a) produced the fraction that proves it. Only a decimal that never ends *and* never settles into a repeating block is irrational. Full credit requires naming the repeating block or citing the fraction from part (a).